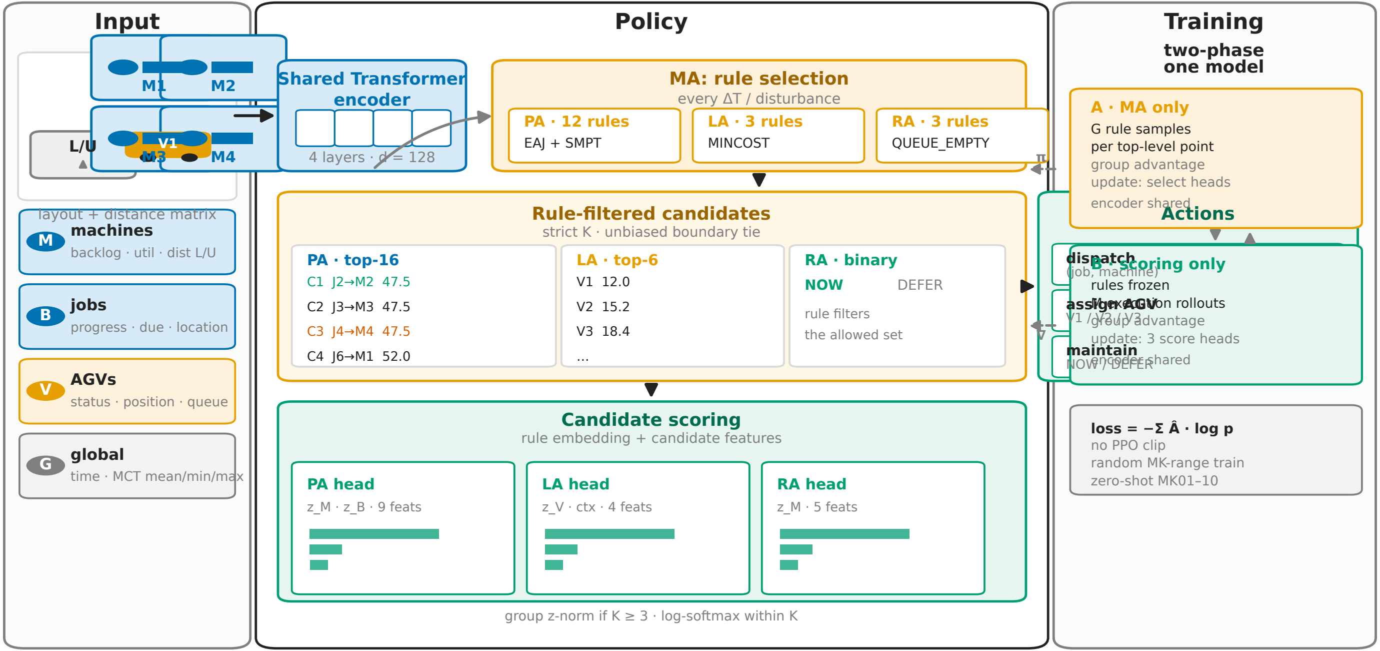
Input
M1
M2
V1
M3
M4
L/U
layout + distance matrix
M machines
backlog · util · dist L/U
B jobs
progress · due · location
V AGVs
status · position · queue
G global
time · MCT mean/min/max
Policy
Shared Transformer encoder
4 layers · d = 128
MA: rule selection
every ΔT / disturbance
PA · 12 rules
EAJ + SMPT
LA · 3 rules
MINCOST
RA · 3 rules
QUEUE_EMPTY
π
Rule-filtered candidates
strict K · unbiased boundary tie
PA · top-16
C1 J2→M2 47.5
C2 J3→M3 47.5
C3 J4→M4 47.5
C4 J6→M1 52.0
LA · top-6
V1 12.0
V2 15.2
V3 18.4
…
RA · binary
NOW
DEFER
rule filters
the allowed set
Candidate scoring
rule embedding + candidate features
PA head
z_M · z_B · 9 feats
LA head
z_V · ctx · 4 feats
RA head
z_M · 5 feats
group z-norm if K ≥ 3 · log-softmax within K
Actions
dispatch
(job, machine)
assign AGV
V1 / V2 / V3
maintain
NOW / DEFER
∇
Training
two-phase
one model
A · MA only
G rule samples
per top-level point
group advantage
update: select heads
encoder shared
B · scoring only
rules frozen
M execution rollouts
group advantage
update: 3 score heads
encoder shared
loss = −Σ Â · log p
no PPO clip
random MK-range train
zero-shot MK01–10